RSA Algorithm in Cryptography

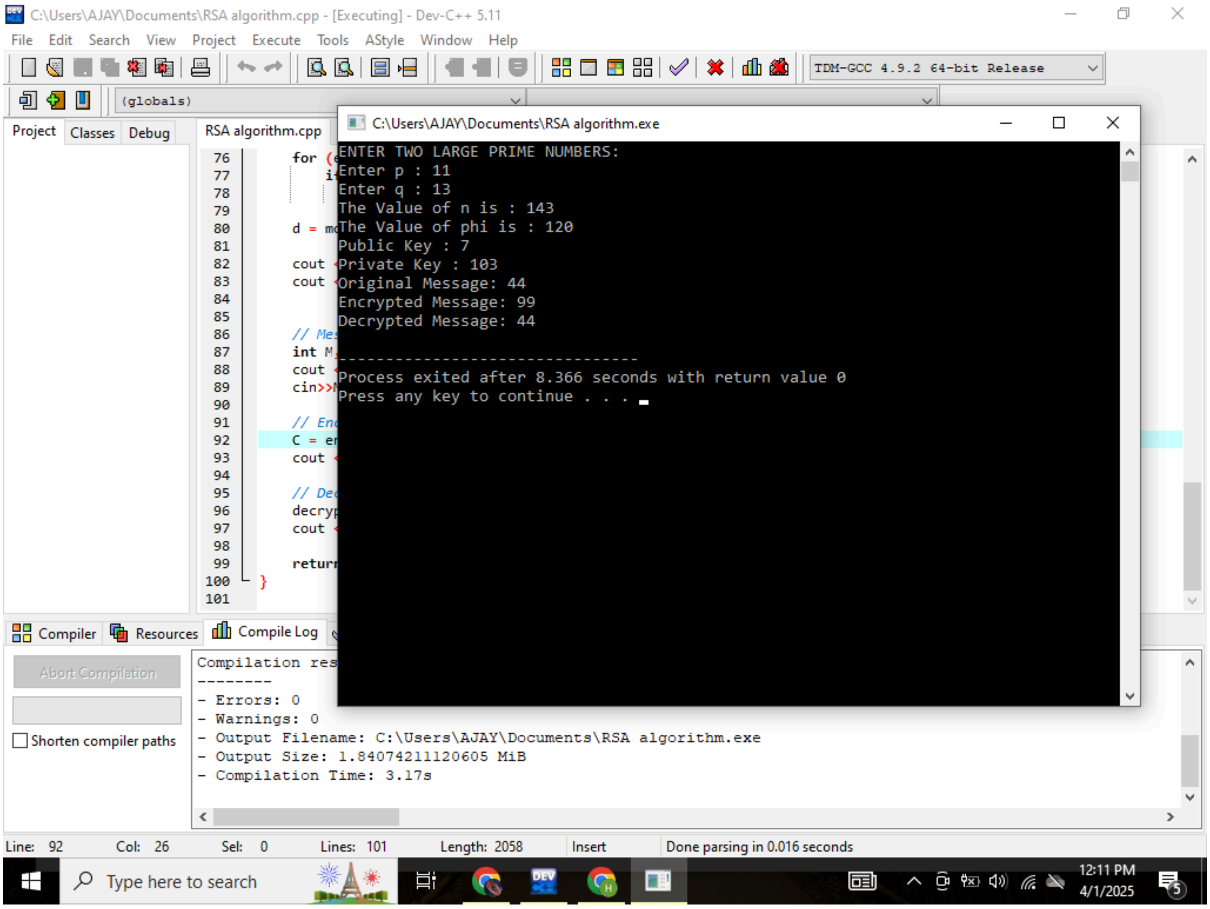
CODE :

```
// C Program for implementation of RSA Algorithm
#include <bits/stdc++.h>
#include<conio.h>

using namespace std;

int power(int base, int expo, int m) {
    int res = 1;
    base = base % m;
    while (expo > 0) {
        if (expo & 1)
            res = (res * 1LL * base) % m;
        base = (base * 1LL * base) % m;
        expo = expo / 2;
    }
    return res;
}

int prime(long int pr)
{
    int i,j;
    j = sqrt(pr);
    for (i = 2; i <= j; i++)
    {
        if (pr % i == 0)
            return 0;
    }
    return 1;
}

int modInverse(int e, int phi) {
    for (int d = 2; d < phi; d++) {
```

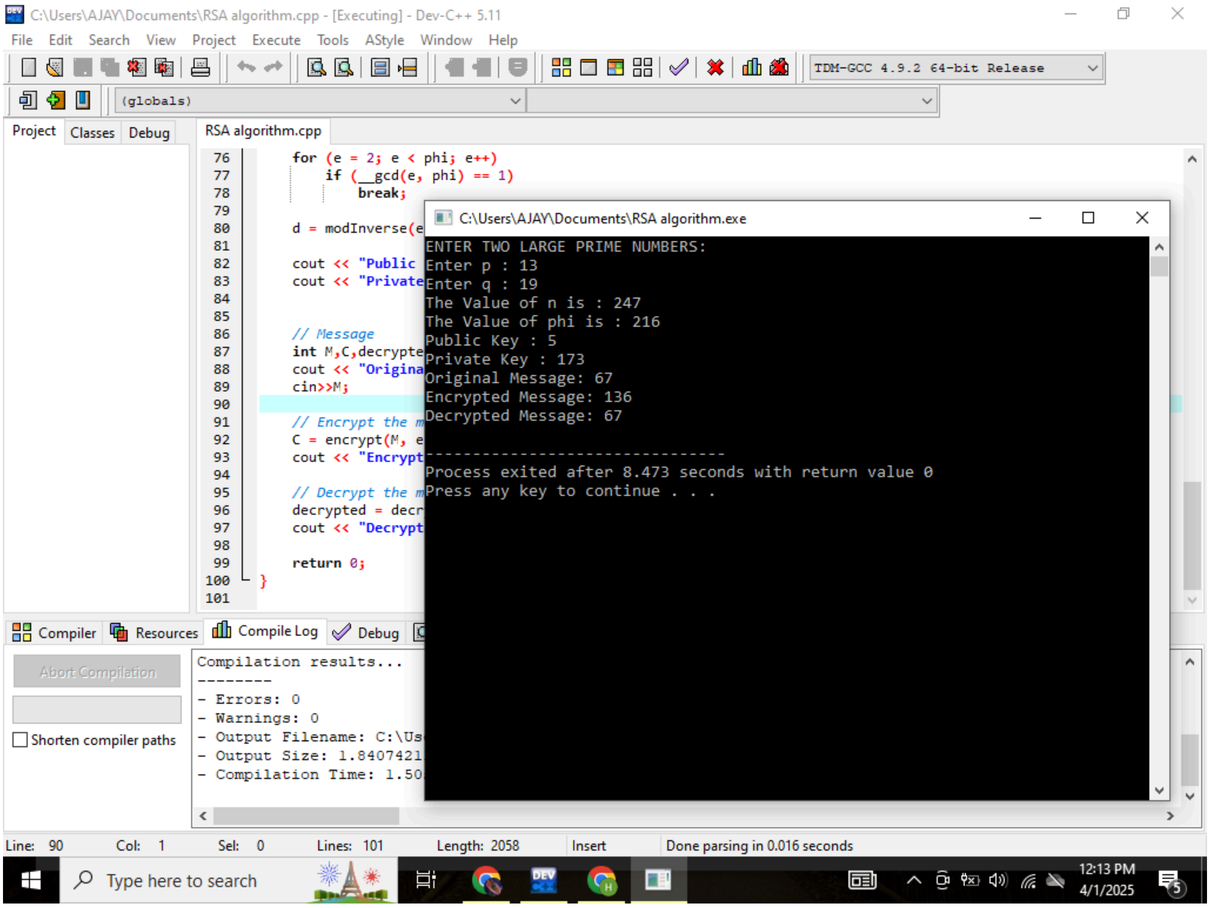
```
        if ((e * d) % phi == 1)
            return d;
    }
    return -1;
}

int encrypt(int m, int e, int n) {
    return power(m, e, n);
}

int decrypt(int c, int d, int n) {
    return power(c, d, n);
}

int main() {
    int e, d, n,p,q,phi;
    bool flag;

    cout << "ENTER TWO LARGE PRIME NUMBERS: "<<endl;
    cout << "Enter p : ";
    cin>>p;
    flag = prime(p);
    if (flag == 0)
    {
        printf("\nWRONG INPUT\n");
        getch();
        exit(1);
    }
    cout << "Enter q : ";
    cin>>q;
    flag = prime(q);
    if (flag == 0 || p == q)
    {
        printf("\nWRONG INPUT\n");
        getch();
        exit(1);
    }
```

```
    n = p * q;
    phi = (p - 1) * (q - 1);
    cout << "The Value of n is : " << n << endl;
    cout << "The Value of phi is : " << phi << endl;
    // Key Generation (e and d)
    for (e = 2; e < phi; e++)
        if (__gcd(e, phi) == 1)
            break;

    d = modInverse(e, phi);

    cout << "Public Key : " << e << endl;
    cout << "Private Key : " << d << endl;


    // Message
    int M,C,decrypted;
    cout << "Original Message: ";
    cin>>M;

    // Encrypt the message
    C = encrypt(M, e, n);
    cout << "Encrypted Message: " << C << endl;

    // Decrypt the message
    decrypted = decrypt(C, d, n);
    cout << "Decrypted Message: " << decrypted << endl;

    return 0;
}
```

OUTPUT :

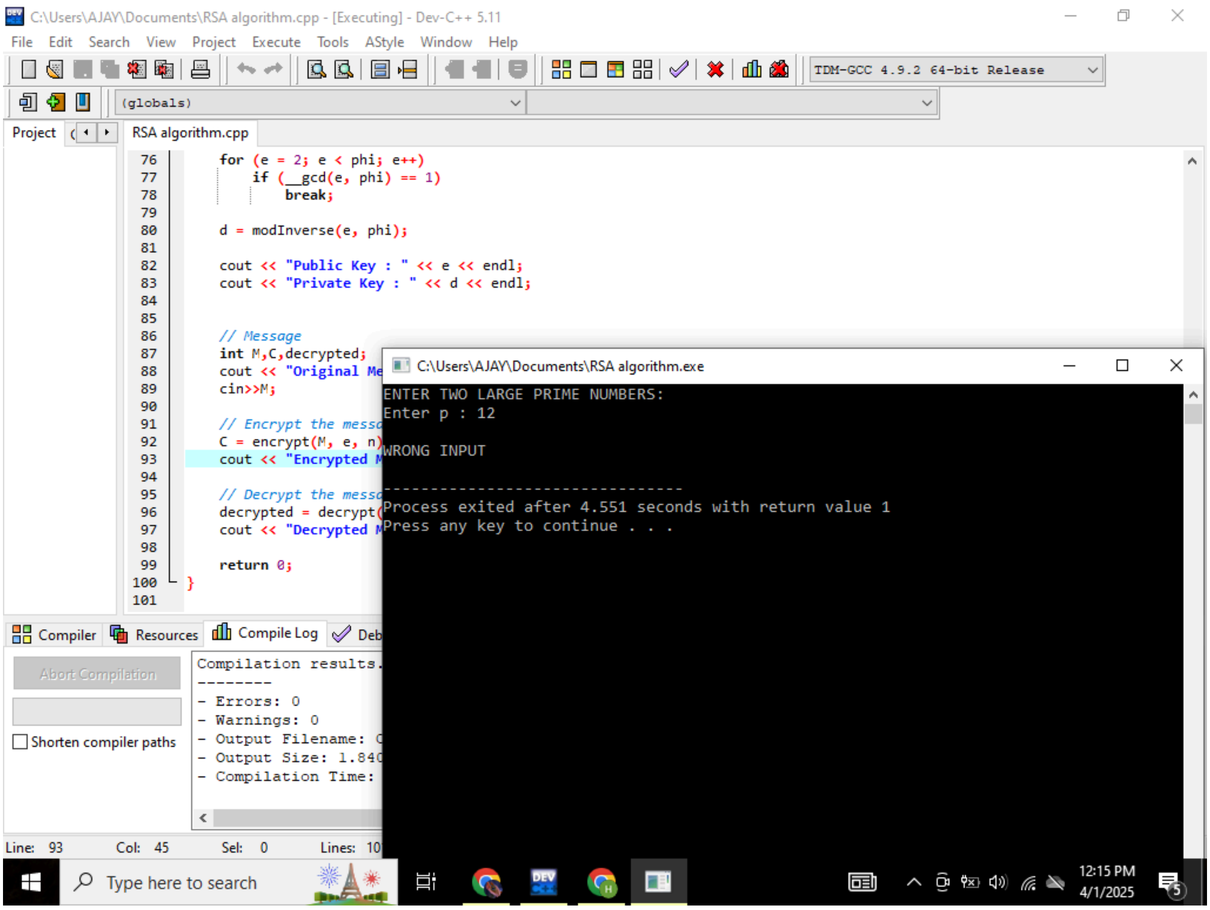

C:\Users\AJAY\Documents\RSA algorithm.cpp - [Executing] - Dev-C++ 5.11
C:\Users\AJAY\Documents\RSA algorithm.exe
ENTER TWO LARGE PRIME NUMBERS:
Enter p : 11
Enter q : 13
The Value of n is : 143
The Value of phi is : 120
Public Key : 7
Private Key : 103
Original Message: 44
Encrypted Message: 99
Decrypted Message: 44

Process exited after 8.366 seconds with return value 0
Press any key to continue . . .
Compilation res
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\AJAY\Documents\RSA algorithm.exe
- Output Size: 1.84074211120605 MiB
- Compilation Time: 3.17s
Line: 92 Col: 26 Sel: 0 Lines: 101 Length: 2058 Insert Done parsing in 0.016 seconds
Type here to search
12:11 PM
4/1/2025


C:\Users\AJAY\Documents\RSA algorithm.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
RSA algorithm.cpp
for (e = 2; e < phi; e++)
if (__gcd(e, phi) == 1)
break;
d = modInverse(e
cout << "Public
cout << "Private
// Message
int M,C,decrypte
cout << "Origina
cin>>M;
// Encrypt the m
C = encrypt(M, e
cout << "Encrypt
// Decrypt the m
decrypted = decr
cout << "Decrypt
return 0;
}
C:\Users\AJAY\Documents\RSA algorithm.exe
ENTER TWO LARGE PRIME NUMBERS:
Enter p : 13
Enter q : 19
The Value of n is : 247
The Value of phi is : 216
Public Key : 5
Private Key : 173
Original Message: 67
Encrypted Message: 136
Decrypted Message: 67

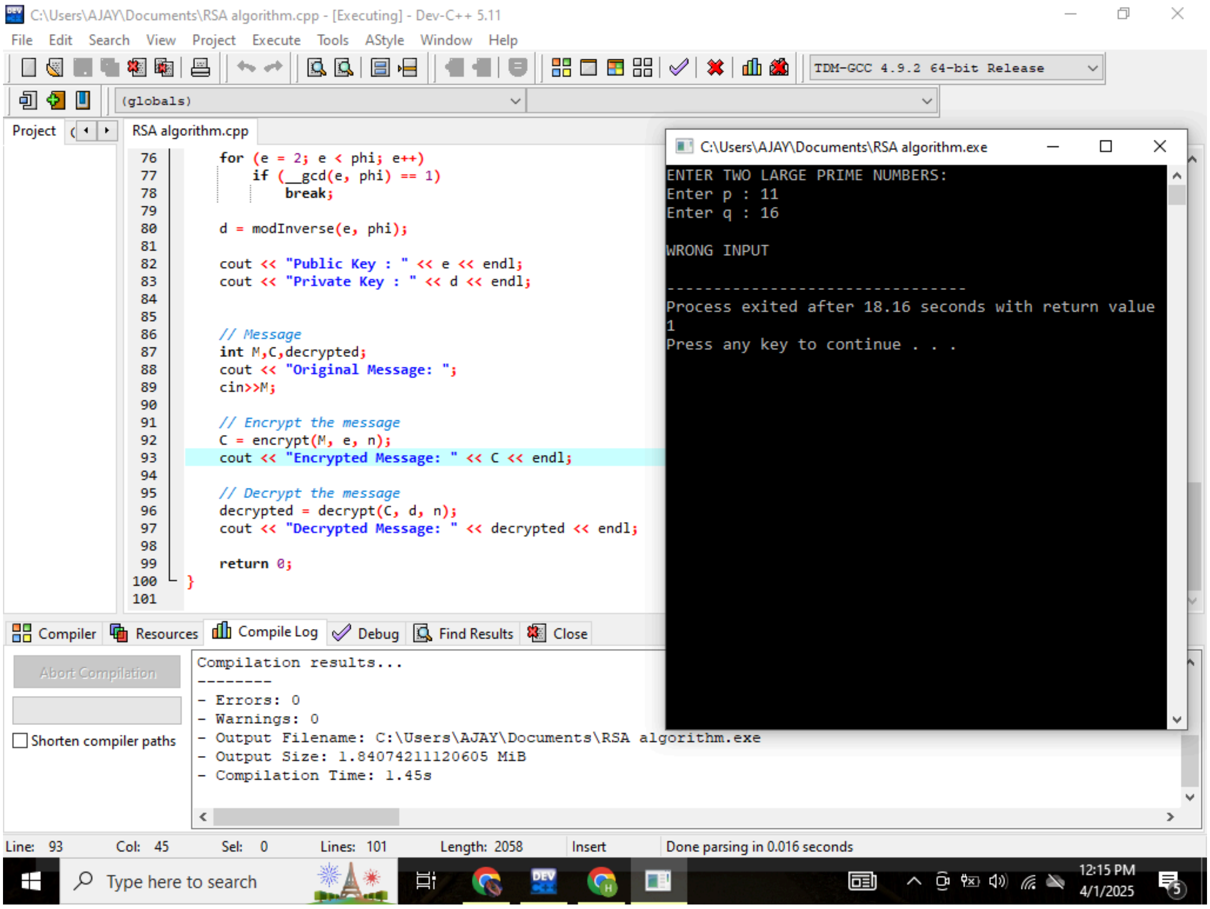
Process exited after 8.473 seconds with return value 0
Press any key to continue . . .
Compiler Resources Compile Log Debug
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Us
- Output Size: 1.8407421
- Compilation Time: 1.50
Line: 90 Col: 1 Sel: 0 Lines: 101 Length: 2058 Insert Done parsing in 0.016 seconds
Type here to search
12:13 PM
4/1/2025


C:\Users\AJAY\Documents\RSA algorithm.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project
RSA algorithm.cpp
for (e = 2; e < phi; e++)
if (__gcd(e, phi) == 1)
break;
d = modInverse(e, phi);
cout << "Public Key : " << e << endl;
cout << "Private Key : " << d << endl;
// Message
int M,C,decrypted;
cout << "Original Me
cin>>M;
// Encrypt the messa
C = encrypt(M, e, n)
cout << "Encrypted M
// Decrypt the messa
decrypted = decrypt(
cout << "Decrypted M
return 0;
}
C:\Users\AJAY\Documents\RSA algorithm.exe
ENTER TWO LARGE PRIME NUMBERS:
Enter p : 12
WRONG INPUT

Process exited after 4.551 seconds with return value 1
Press any key to continue . . .
Compiler Resources Compile Log Deb
Abort Compilation
Shorten compiler paths
Compilation results.

- Errors: 0
- Warnings: 0
- Output Filename: C
- Output Size: 1.840
- Compilation Time:
Line: 93 Col: 45 Sel: 0 Lines: 10
Type here to search
12:15 PM
4/1/2025


C:\Users\AJAY\Documents\RSA algorithm.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project
RSA algorithm.cpp
for (e = 2; e < phi; e++)
if (__gcd(e, phi) == 1)
break;
d = modInverse(e, phi);
cout << "Public Key : " << e << endl;
cout << "Private Key : " << d << endl;
// Message
int M,C,decrypted;
cout << "Original Message: ";
cin>>M;
// Encrypt the message
C = encrypt(M, e, n);
cout << "Encrypted Message: " << C << endl;
// Decrypt the message
decrypted = decrypt(C, d, n);
cout << "Decrypted Message: " << decrypted << endl;
return 0;
}
C:\Users\AJAY\Documents\RSA algorithm.exe
ENTER TWO LARGE PRIME NUMBERS:
Enter p : 11
Enter q : 16
WRONG INPUT

Process exited after 18.16 seconds with return value 1
Press any key to continue . . .
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\AJAY\Documents\RSA algorithm.exe
- Output Size: 1.84074211120605 MiB
- Compilation Time: 1.45s
Line: 93 Col: 45 Sel: 0 Lines: 101 Length: 2058 Insert Done parsing in 0.016 seconds
Type here to search
12:15 PM
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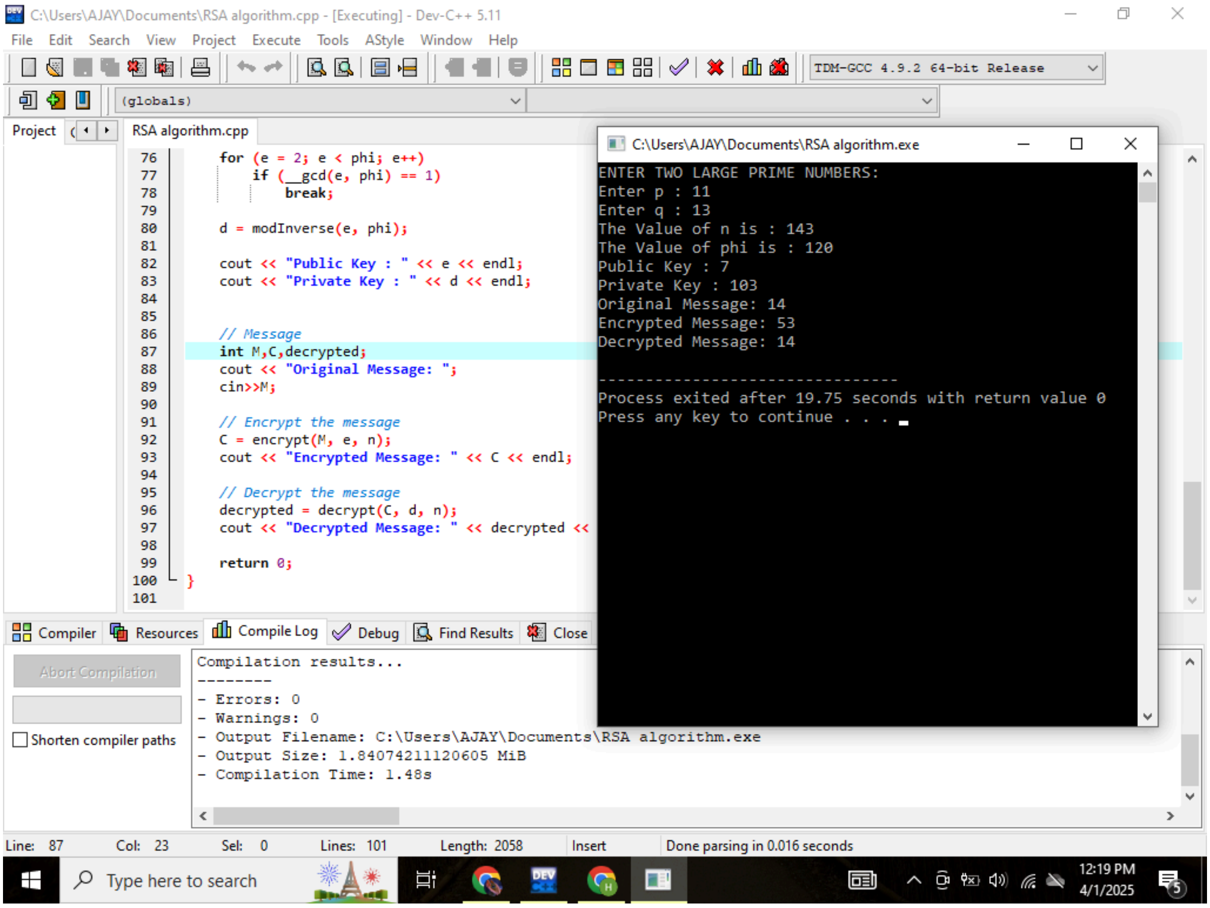
C:\Users\AJAY\Documents\RSA algorithm.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project
RSA algorithm.cpp
for (e = 2; e < phi; e++)
if (__gcd(e, phi) == 1)
break;
d = modInverse(e, phi);
cout << "Public Key : " << e << endl;
cout << "Private Key : " << d << endl;
// Message
int M,C,decrypted;
cout << "Original Message: ";
cin>>M;
// Encrypt the message
C = encrypt(M, e, n);
cout << "Encrypted Message: " << C << endl;
// Decrypt the message
decrypted = decrypt(C, d, n);
cout << "Decrypted Message: " << decrypted <<
return 0;
}
C:\Users\AJAY\Documents\RSA algorithm.exe
ENTER TWO LARGE PRIME NUMBERS:
Enter p : 11
Enter q : 13
The Value of n is : 143
The Value of phi is : 120
Public Key : 7
Private Key : 103
Original Message: 14
Encrypted Message: 53
Decrypted Message: 14

Process exited after 19.75 seconds with return value 0
Press any key to continue . . .
Compiler
Resources
Compile Log
Debug
Find Results
Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\AJAY\Documents\RSA algorithm.exe
- Output Size: 1.84074211120605 MiB
- Compilation Time: 1.48s
Line: 87
Col: 23
Sel: 0
Lines: 101
Length: 2058
Insert
Done parsing in 0.016 seconds
Type here to search
12:19 PM
4/1/2025